

Package #107

Standardize thread geometry and material composition across all components in the pedicle screw system

Technology Maturity: ENGINEERING_DEFINITION

Transfer Posture: Sponsored Validation

Problem: Documented recurring problem 'break' for pedicle screw: 11 MAUDE reports across 3 distinct devices (20080306 to 20201006); root-cause evidence from 20 recall records and 10 event narratives (custody attached in the campaign report). MAUDE limitations: incidence UNKNOWN, causation UNVERIFIED (Art. XXI.5). Existing solutions fail because: Why existing solutions fail, per retrieved evidence: 20 recall root-cause descriptions (FDA enforcement records, custody attached); 10 MAUDE event narratives from this mechanism cluster (mdr_text, custody attached). Statements are the providers' own words, cus Unresolved gap: Across the 6 relevant retrieved attempts (a bounded, non-exhaustive set), none measurably addresses: break, conditions, devices, events, failure, function, intervention, introducing, maintain, malfunction, matching, modes, must, problem, producing, recurring,

Mechanism: Device-device compatibility issues leading to mechanical stress concentrations

Contents:

- 01 - Executive Technology Brief
- 02 - Engineering Technology-Transfer Dossier
- 03 - Buyer Decision Card
- 04 - Evidence Summary
- 05 - Transfer Manifest

Status: ENGINEERING_DEFINITION. Not physically validated.